
STEEL DEFECT DETECTION

PREZENTARE MIDTERM

Scopul proiectului

Cercetari anterioare

Structura datelor

Modele

Evaluare (Metrics)

Metode de optimizare

Rezultate si recomandari

SCOPUL PROIECTULUI

- Localizarea si clasficarea defectelor pe suprafete metalice
- La final, dorim sa prezentam o gama de modele care poate verifica si clasifica kilometrii de benzi de metalurgii si vom specifica ce model te va ajuta cel mai mult in functie de resursele tale.

CERCETARI ANTERIOARE

- [https://medium.com/@smv9495/severstal-steel-defect-detection-1b5966dc0512#:~:text=This%20is%20a%20Kaggle%20competition,%2C%203%2C%204%5D%20\).](https://medium.com/@smv9495/severstal-steel-defect-detection-1b5966dc0512#:~:text=This%20is%20a%20Kaggle%20competition,%2C%203%2C%204%5D%20).) (Severstal Steel Defect Detection)
 - <https://link.springer.com/article/10.1007/s42979-023-02436-2> (A Review and Benchmark on State-of-the-Art Steel Defects Detection)
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STRUCTURA DATELOR

- Severstal: Steel Defect Detection (Kaggle) 12.6k - Training | 5506 - Testing

Fiecare imagine poate avea un defect, mai multe defecte, sau niciunul (defectele sunt segmentate in clasa "ClassId = [1, 2, 3, 4]").

Segmentul fiecare clase este encodat intr-un singur rand.

- NEU surface defect database 1,440 - Training | 360 - Testing

// UNDER CONSTRUCTION //

MODELE

1. U-Net - "U-shaped" encoder-decoder structure with skip connections
2. Mask R-CNN - Region-based CNN which divides object detection task into two stages: region proposal and object classification
3. YOLOv11- Popular object detection & image segmentation model
4. Vit - Deep learning architecture that applies the Transformer model to images
5. Sam – Model which produces high quality object masks from input prompts such as points or boxes, used to generate masks for all objects in an image.
6. Dyno - A Self-supervised Vision Transformer Model

EVALUARE (METRICS)

- Intersection over Union (IoU)

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} = \frac{TP}{TP + FP + FN}$$

- Dice Coefficient

$$Dice = \frac{2 \times \text{Area of Overlap}}{\text{Total number of pixels in both}} = \frac{2TP}{2TP + FP + FN}$$

- Pixel Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

METODE DE OPTIMIZARE

- **// UNDER CONSTRUCTION //**

REZULTATE SI RECOMANDARI

- **// UNDER CONSTRUCTION //**

MAI DEPARTE

- Implementare cod
- Optimizari pentru imbunatatirea modelelor
- Vizualizari de predictii